Udyam Digital — solution architecture

Accel Business · digital acquisition & onboarding for MSMEs · everything below runs today in the prototype; dashed boxes are production integrations

OWNER'S PHONE (BROWSER)

① Advisor tap · type · speak (EN / हिं / मरा)
free-text parser + live bundle, instant on device

② Quote & checkout
one price · add-on toggles · contract · dependencies

③ Smart KYC on-device
consent → blur check → auto-straighten →
Tesseract OCR (WASM) → field extraction
images never leave the phone

④ Order tracker
live stages · install slot · tickets · no calls

⑤ WhatsApp-style chat + voice
same advisor over a webhook contract

Shared rules engine recommend · price · parse

⑥ Partner & sales console
hand-offs from any screen · call list ranked by propensity (AUC 0.71) · funnel insights · audit log
assign to inside sales / channel partner / field

profile

lines

fields + quality only

poll 5 s

ACCEL BACKEND · NEXT.JS ROUTE HANDLERS

/api/advisor/parse · /explain
rules first; AI refines & explains (3.5 s timeout → template)

/api/gstin/:id · /api/udyam/:no mock registry APIs
format + check digit + status; returns only needed fields

/api/kyc/verify server re-validates
GSTIN checksum + OCR-confusion repair · registry match · names · address · bill ≤ 90 days · signatory · verdict

/api/consent
per purpose · versioned

/api/orders
re-prices, re-checks rules

/api/orders/:id
status from time · slots

/api/leads
hand-off + propensity

/api/whatsapp/webhook
{from, text | optionId} → text / buttons / list replies

/api/voice/stt · /tts · /api/kyc/extract
optional AI paths; 501 → browser speech / on-device OCR

AI router
en → NVIDIA NIM (chat) · hi/mr → Sarvam
AI never picks products or prices — it only explains
every call: hard timeout + deterministic fallback

masked

AI · DATA · INTEGRATIONS

NVIDIA NIM
chat: deepseek-v4.1-flash / nemotron-3.5
vision: nemotron-3.5-super-vl (doc reading)

Sarvam
Hindi/Marathi replies · Saarika STT · Bulbul TTS

SQLite (node:sqlite)
orders · consents · leads · tickets · chat sessions
events = audit trail with masked identifiers
no document images stored anywhere
no PII sent to AI providers (GSTIN stripped)

Data pack → JSON (build step)
catalogue (25) · GSTIN registry (3,534)
Udyam registry (2,723) · cities → tier
analyze.py → insights · propensity · call list

Production integrations (swap mocks)
GSTN / Udyam APIs (via GSP) · DigiLocker
Aadhaar e-Sign (ESP) · UPI AutoPay mandate
WhatsApp Cloud API (thin adapter)
OSS/BSS provisioning → tracker events
Fibre feasibility API → AirFiber fallback
CRM / partner app for hand-offs

Why it never hangs
Bundle, prices and parsing run instantly in the browser · pages are pre-rendered static shells · AI text streams token by token · OCR worker is self-hosted and warmed up on the consent screen